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Time-Phased Relief Supply Network Planning for Foreseen Disasters

Vala Rahmati,[a] Halit Üster[a,*]

[a] Department of Operations Research and Engineering Management, Lyle School of Engineering, Southern Methodist University, Dallas, Texas 75275

*Corresponding author

Contact: vrahmati@mail.smu.edu, https://orcid.org/0009-0009-1942-2231 (VR); uster@smu.edu, https://orcid.org/0000-0002-6714-5402 (HÜ)

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Abstract. We consider emergency preparedness for foreseen disasters such as hurricanes and flooding. Proper preparation and planning for timely relief supply in a cost-effective manner can be crucial determinants of how quickly recoveries occur and with the least suffering of the affected populations. Assuming a host of shelter locations aggregated locally, we are interested in determining relief supply locations (distribution centers [DCs]) and routing of supply from the capacitated DCs to the shelters on an underlying time-phased network for cost-effective timely delivery. We present a mixed integer optimization model to address this problem under the assumption of covering the worst case demand at the shelters. To solve our model, we develop an efficient Benders decomposition–based algorithm that handles the challenges of obtaining optimality cuts via master problem solution modifications and various surrogate constraints among other enhancement techniques. We test the performance of the enhancement techniques on an extensive randomly generated test data set to identify the most effective approach. We finally use our model and the solution algorithm on an actual case study in southern Texas data incorporated and managed by a geographical information system to examine the impact of various input parameters on the design and tactical operation of the relief networks as well as for further model verification and validation.

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1. Introduction

Considering global population growth, there is a significantly increasing possibility of exposure to natural disasters (e.g., hurricanes, flooding, etc.), making them a real threat to human beings (Schumacher and Strobl 2011). According to CBS News (2024), Hurricane Maria was the most destructive hurricane in the United States since 2000. It originated in the Indian Ocean and then moved westward across the Atlantic Ocean, causing 2,975 deaths and approximately $90 billion in property damage in 2017. Hurricanes such as Katrina in 2005, Harvey in 2017, and Ian in 2022, among other natural disasters, highlight how these events can be devastating and result in the loss of human lives. Growing concerns that are also fueled by increasingly rapid intensification of tropical cyclones to category 3–5 hurricanes and ensuing major flooding are outlined based on past data (Climate Central 2025). Responding to such disasters requires preplanned and well-organized actions to enhance the efficiency and effectiveness of the responses. Therefore, authorities strive to develop plans that support the affected population before, during, and after such disasters.

In light of these threats, the preparedness and response stages in disaster management require careful coordination of both strategic and tactical decisions. In the context of humanitarian logistics, strategic decisions involve establishing the underlying physical network, such as selecting which distribution centers (DCs) to open and which transportation links or transfer nodes to activate prior to or immediately following the disaster onset. Tactical decisions, on the other hand, prepare for the operational execution within that activated network, focusing on the time-dependent routing, scheduling, and allocation of relief supplies to shelters (Üster and Dalal 2017).

Traditionally, these decisions are often treated sequentially or independently; however, in large-scale events, this separation can lead to operational inefficiencies. Activating infrastructure without accounting for time-dependent routing dynamics may create bottlenecks when multiple supply flows are forced through limited-capacity road segments or transfer nodes at the same time, resulting in delayed deliveries and infeasible relief dispatch schedules even when sufficient total supply exists. For example, opening a limited set of DCs or transportation links may concentrate flows along a small number of corridors, causing capacity conflicts that prevent supplies from reaching shelters within required time windows. Conversely, planning routing decisions on a rigid, predetermined network limits response capacity because alternative facilities or transportation links cannot be activated when accessibility constraints, infrastructure damage, or demand surges make existing routes insufficient. As a result, operational plans may remain feasible only on paper, failing to meet delivery timing requirements in practice. Integrating strategic (activation) and tactical (routing and scheduling) decisions within a multiperiod framework, therefore, becomes essential. This integration allows decision makers to navigate the core tradeoff of relief logistics: balancing the fixed costs of activating additional infrastructure against the operational necessity of delivering supplies early enough to satisfy shelter demand and avoid capacity-driven delays.

In this study, we consider time-expanded relief distribution network design to optimize the prompt delivery of various supplies from DCs to shelters through transfer nodes in a geographical area. Thus, our designed network comprises a three-tier system, including DCs, transfer nodes, and shelters. Transfer nodes (points) are the nodes of the underlying road network through which relief supply can flow on their path from origins (DCs) to destinations (shelters). We assume that all shelters planned to be open must be ready to run at their maximum capacity and be fully supplied.

To connect the model structure with real-world operations—such as those observed during major hurricanes along the Texas Gulf Coast (Federal Emergency Management Agency 2018)—the network is constructed from geographical information system (GIS)–based representations of the transportation system used in disaster response planning. Candidate DCs correspond to cities with relatively higher population levels that are used as proxy supply or staging locations, whereas shelters correspond to school facilities that have previously served as evacuation shelters in historical disaster events. Transfer nodes represent intersections and intermediate points on the underlying road network through which relief traffic must pass. Travel times and accessibility conditions are derived from preprocessing of the road network using distance, road capacity, and expected usage patterns, producing fixed travel-time parameters that approximate postdisaster operating conditions, maintaining computational tractability.

Because travel times are treated as fixed inputs, the model does not capture endogenous congestion effects in which routing decisions dynamically alter travel times. Instead, operational difficulty associated with capacity-limited or structurally important parts of the network is represented indirectly through preprocessing-based travel-time adjustments, explicit arc-capacity constraints, and population-dependent operational costs at transfer nodes. Preprocessed travel times reflect expected operating conditions, whereas arc-capacity constraints limit the feasible throughput of relief flow within each time period and do not introduce additional congestion penalties. These parameters reflect expected operational limitations in densely populated areas and encourage routing decisions that avoid highly constrained parts of the network without introducing flow-dependent feedback mechanisms. From a tactical perspective, our time-expanded network manages aggregate flows across different time periods. In application, this approach enables adjustment of relief flows according to the timing of evacuations directed toward shelters.

Whereas this integrated approach provides valuable insights into decision making for timely, adequate, and cost-effective delivery of relief supplies to evacuee demand points,[1] the complexities inherent in our problem and the dynamics of a time-expanded network make the corresponding optimization model challenging to solve with many decisions constrained by various factors and nuances. To navigate this issue, we devise an enhanced Benders decomposition (BD) algorithm with algorithmic features that can also be useful in other time-expanded network design settings. BD is a technique known for its effectiveness in breaking down large-scale mixed-integer linear programming (LP) problems into more manageable master and subproblem(s), which are solved iteratively. In our case, the master problem corresponds to obtaining the strategic underlying network structure (activating facilities and links), and the subproblem evaluates the tactical network flow on that time-phased network.

The inherent flexibility of BD provides opportunity to address improving upper and lower bounds in an iterative scheme; however, as is usually the case, a standard application of BD alone proves insufficient for achieving an efficient approach. One reason for this, as is the case for our problem, can be due to the weakness of the Benders (infeasibility) cuts in the cases in which the subproblem feasibility cannot be guaranteed by any master problem solution. This is, in a way, the curse of decomposition that separates the constraints between the subproblem and the master subproblem that becomes oblivious to some of the important problem features expressed by the constraints that are now in the subproblem. In turn, this leads to weak lower bounds provided by the master problem and infeasibility of the subproblem(s) for fixed master problem solutions among other difficulties. Nevertheless, decomposition also provides an opportunity for substantial improvements in the bounds by expediting the convergence toward a near-optimal solution.

To this end, for our problem at hand, we introduce acceleration techniques that include cut strengthening for improved lower bounds, an alternative formulation for separability of the subproblem, an ϵ-optimal method to reach faster a master problem solution, and surrogate constraints for feasibility of the subproblems and, thus, to obtain reasonable upper bounds and Benders optimality cuts rather than weaker feasibility cuts. Furthermore, our preliminary computational study shows that most of the effort is spent on reaching an optimal master problem solution that provides a feasible primal subproblem solution for optimality Benders cuts (rather than feasibility cuts). Noticing that the subproblem is infeasible for the given master problem solution mainly because of capacity restrictions and an unconnected underlying network, we devise an algorithm that provides an optimal master problem solution with carefully selected additional arcs for a connected network. This is achieved by solving an associated max flow problem and utilizing zero residual arcs and a reduced network design problem. Our approach significantly enhances the overall efficiency of the solution process.

To test and validate our proposed method, we conduct numerical studies across three different network sizes and eight distinct classes of problem instances. Additionally, we apply our model and our proposed algorithm to actual data from the Texas Gulf Coast as a case study by employing a GIS. We further explore the influence of input parameters on the optimal solution by conducting sensitivity analysis, thereby examining the relationship between model parameters and the optimal outcome.

The remainder of this study is organized as follows. In Section 2, we review the most related literature to our problem. In Section 3, we outline the problem statement and its assumptions followed by, in Section 4, the introduction of notation and the mathematical model. The proposed solution methodology based on Benders decomposition and acceleration strategies are detailed in Section 5. In Section 6, we summarize the computational study evaluating the performance of the solution approach and its components with full details provided in the online supplement. In Section 7, we present a GIS-based case study using real data from the Texas Gulf Coast to validate and illustrate the model with sensitivity analyses provided in the online supplement. Finally, in Section 8, we provide concluding remarks and directions for future research.

2. Related Literature

Numerous studies examine the design and operation of relief networks in disaster management with optimization-based models commonly aiming to minimize operational costs; reduce delivery times for critical supplies; and improve service equity under practical constraints, such as limited budgets, facility and transportation capacities, and accessibility disruptions. Comprehensive surveys, including Galindo and Batta (2013), summarize methodological developments in humanitarian logistics and relief network design. Emergency logistics research spans the full disaster cycle from preparedness planning to postdisaster response operations (Kundu, Sheu, and Kuo 2022), and existing models differ primarily in the scope of decisions considered and in how operations are represented over time. Strategic preparedness models typically focus on infrastructure siting and inventory prepositioning before a disaster, whereas tactical response models emphasize routing, allocation, and distribution decisions after an event. Accordingly, the literature is organized based on these decision structures and their treatment of time and flow representation to clarify the contribution of the integrated strategic–tactical framework developed in this study.

Early strategic preparedness studies in relief supply logistics primarily focus on improving response efficiency once a disaster occurs. A central theme in this stream is determining where facilities should be located and how relief resources should be positioned prior to an event so that affected regions can be served more rapidly during the response phase. For example, Duran, Gutierrez, and Keskinocak (2011) develop a global prepositioning framework for CARE International that evaluates warehouse locations and replenishment policies across disaster-prone regions, showing that a limited number of strategically located facilities can substantially reduce average response times. Similarly, hierarchical facility planning models proposed by Widener and Horner (2011) and Horner and Downs (2010) consider multitier distribution structures in which higher level facilities provide backup coverage for lower level sites, improving spatial accessibility and system robustness under anticipated demand patterns. In a related direction, Ukkusuri and Yushimito (2008) incorporate inventory prepositioning decisions within a facility location framework that accounts for transportation feasibility and potential network disruptions during hurricane response planning. Similar preparedness-oriented formulations incorporate uncertainty in robust or min–max frameworks in which facility locations and inventory levels are determined to ensure feasibility across disruption scenarios. Postdisaster distribution decisions are represented at an aggregate level rather than through time-dependent operational flows (Ni, Shu, and Song 2018).

Subsequent strategic planning studies extend this preparedness perspective by also incorporating operational considerations. Rodríguez-Espíndola, Albores, and Brewster (2018) combine GIS-based hazard mapping with location and resource allocation decisions to support coordinated planning across multiple agencies. Protogyrou and Hajibabai (2024) introduce accessibility-based prioritization and multimodal transportation options to capture service priorities in flood response. In these models, distribution decisions are typically represented at an aggregate level and operational routing and/or dispatch timing are not explicitly modeled. In addition, infrastructure and network representations are typically aggregated, whereas operational relief planning often relies on GIS-derived road networks with designated shelter locations and region-specific accessibility assumptions. As a result, once facility and inventory decisions are determined, the subsequent movement of relief supplies is generally assumed to occur within a preset period rather than evolving over time across the network. In summary, whereas strategic preparedness models provide important insights into infrastructure and resource planning, they do not explicitly represent how relief should be planned to flow through an activated network in the response stage. In particular, operational decisions such as dispatch timing and routing of relief supplies and the interaction between infrastructure activation and delivery feasibility are typically not addressed in these studies. Such limitations necessitate research that focuses on tactical and operational decision making for the response stage for which routing and allocation decisions are modeled explicitly in a way to examine routing feasibility changes when infrastructure activation decisions are determined simultaneously with operational planning.

In the response stage, research primarily focuses on tactical and operational decisions, including routing; allocation; and the timely delivery of relief supplies under capacity, accessibility, and equity considerations. In contrast to preparedness-oriented models, these studies assume that the underlying infrastructure is already determined and concentrate on improving operational efficiency within this fixed network structure. For example, Eisenhandler and Tzur (2019a) introduce the humanitarian pickup and distribution problem, which models equitable collection and distribution activities using limited-capacity vehicles. Eisenhandler and Tzur (2019b) improve computational tractability through a segment-based formulation that preserves allocation fairness. Mostajabdaveh, Salman, and Gutjahr (2025) address last mile distribution to jointly optimize vehicle routes and delivery quantities from a central depot location to shelters, and Beheshtinia, Jozi, and Fathi (2023) consider heterogeneous fleets in a setting in which the routing feasibility and delivery performance are dependent on vehicle compatibility and warehouse capacities.

A related stream extends routing models by also incorporating other logistical components, including production planning, inventory decisions, or rolling-horizon considerations. Zargary and Samouei (2022) propose an integrated production–routing–inventory formulation that coordinates manufacturing, storage, and delivery decisions in a multiperiod setting. Rivera-Royero, Galindo, and Yie-Pinedo (2020) integrate relief kit assembly, workforce planning, and routing decisions within a rolling-horizon framework. Wang and Nie (2023) consider dynamic distribution planning coordinating location, inventory, and routing decisions over time to improve responsiveness, and Iloglu and Albert (2018) incorporate infrastructure restoration and service scheduling decisions during emergency response. More recently, Gong et al. (2025) emphasize multimodal coordination, such as the integration of trucks, boats, and drones to improve delivery efficiency under disruption conditions.

We observe that, in the studies that even consider multiperiod or rolling-horizon settings, transportation of relief supplies is typically represented via vehicle routes and aggregated shipments rather than time-indexed flows in the network (Iloglu and Albert 2018; Rivera-Royero, Galindo, and Yie-Pinedo 2020; Zargary and Samouei 2022). Consequently, routing and delivery decisions are optimized independently of infrastructure activation decisions, and the interaction between opening facilities or transportation links and the resulting operational feasibility is not explicitly captured. These limitations motivate the development of integrated models that jointly consider strategic and operational decisions, and these are discussed next.

Finally, a complementary stream of research integrates strategic planning and tactical/operational decisions, explicitly accounting for uncertainty in relief supply logistics. These studies recognize that facility location, inventory positioning, and distribution decisions are interdependent and, therefore, model them jointly within stochastic or robust optimization frameworks. For example, Sheu and Pan (2014) propose a centralized emergency supply network that determines facility locations and distribution decisions under multiple cost considerations, whereas Noyan, Balcik, and Atakan (2016) develop a stochastic programming approach for relief network design that simultaneously optimizes distribution point locations and capacities, maintaining equitable access. Integrated logistics formulations that jointly consider allocation and distribution decisions are also proposed in predictable disaster settings; for example, integrated planning models such as Vanajakumari, Kumar, and Gupta (2016) coordinate facility and distribution decisions within a unified framework but represent distribution through aggregated or route-based decisions without explicitly modeling time-dependent flow propagation through the network. Similarly, Dalal and Üster (2018) present an integrated response network design model that balances worst case and average case objectives, jointly determining supply center and shelter decisions under disaster uncertainty using a Benders decomposition framework. Extensions of this line of work incorporate richer uncertainty representations and multilayer network structures. For instance, Dalal and Üster (2021) model evacuation-side uncertainty within a multitier relief network, capturing interactions between evacuation behavior and supply logistics, whereas Guo, Dong, and Zhu (2025) develop a nested logic-based Benders decomposition approach that coordinates siting, stocking, and routing decisions through a two-stage stochastic formulation. Related studies consider preparedness decisions under uncertainty through robust or scenario-based optimization, including facility location and inventory allocation models that balance cost, coverage, and service reliability across possible disaster scenarios (Erbeyoğlu and Bilge 2020, Hasani and Sheikh 2023, Shu et al. 2023).

Although these integrative studies represent an important step toward strategic and tactical/operational decision making, existing formulations typically consider siting or activation decisions at a single infrastructure level (e.g., warehouses or distribution facilities), whereas operational routing and allocation decisions are performed within a fixed network structure (Sheu and Pan 2014, Dalal and Üster 2018, Erbeyoğlu and Bilge 2020, Hasani and Sheikh 2023, Shu et al. 2023). In contrast, multilevel activation decisions across facilities and transportation links remain limited in the literature. Time is typically represented through stage- or scenario-based structures in which shipments occur within a planning period rather than propagating explicitly through the network over time. As a result, relief distribution is generally modeled through aggregated flows, assignment decisions, or inventory transitions, and operational dispatch timing is not explicitly represented at the arc level (Sheu and Pan 2014; Noyan, Balcik, and Atakan 2016; Dalal and Üster 2018; Guo, Dong, and Zhu 2025). Consequently, whereas infrastructure and allocation decisions are optimized jointly, the interaction between infrastructure activation and the time-dependent feasibility of routing and delivery decisions remain implicit. In particular, existing integrated models do not explicitly capture how relief flows evolve through an activated network across time periods at the arc level or how early dispatch decisions influence downstream capacity utilization and delivery performance.

Taken together with the limitations identified above in preparedness- and response-oriented strategic and tactical/operational routing models, our observations highlight a remaining gap in the literature. Strategic models determine infrastructure and prepositioning decisions but abstract from operational delivery dynamics, whereas tactical routing models optimize distribution within fixed networks. Integrated stochastic and robust formulations combine these decision layers but typically rely on aggregated or stage-based representations of flow. As a result, these formulations generally cannot explicitly evaluate whether a selected infrastructure configuration remains operationally feasible once time-dependent routing and capacity interactions are considered simultaneously. This motivates the development of models that jointly determine infrastructure activation and operational routing decisions, explicitly representing the temporal evolution of relief flows through the network under fixed travel-time assumptions derived from preprocessing, preserving tractability for large-scale disaster networks. The present study addresses this need by introducing a time-expanded relief network that integrates strategic and tactical decisions within a unified framework, allowing the movement of relief supplies to be modeled explicitly over time at the network arc level in large-scale disaster response settings.

2.1. Our Contributions

To address the limitations identified above, this study proposes a time-expanded network designed for the relief phase of large-scale, foreseen disasters, such as hurricanes and floods. The proposed model integrates strategic and tactical decisions within a unified optimization framework. At the strategic level, the model determines the activation of facilities and transportation links within a multitier relief network. At the tactical level, it determines the routing, allocation, and timing of relief flows over the planning horizon. By introducing time-indexed flow variables, the formulation explicitly represents how relief supplies propagate through the network across time periods at the arc level, allowing decision makers to evaluate the feasibility of distribution plans and identify operational bottlenecks arising from capacity and timing constraints.

Travel times are treated as fixed input parameters derived from preprocessing steps that account for baseline accessibility conditions and expected road usage patterns. Rather than modeling endogenous traffic congestion explicitly, the proposed model represents operational difficulty through population-dependent operational costs at transfer and shelter nodes. Higher demand nodes incur higher operational penalties, encouraging routing decisions that avoid heavily utilized or operationally constrained areas without introducing flow-dependent feedback mechanisms. In addition, the formulation introduces a worst case fulfillment requirement at shelters to ensure minimum service levels during widespread evacuations, strengthening the reliability of relief operations under high-demand conditions. Together, these modeling elements provide a transparent representation of the trade-off between infrastructure activation decisions and the timely delivery of relief supplies.

On the algorithmic side, we develop enhanced BD approaches tailored to the structure of the proposed time-expanded network. Both a customized implementation and an automatic CPLEX-based decomposition are developed and benchmarked against a branch-and-cut (B&C) approach. To improve convergence and reduce infeasibility in the subproblems, we introduce an auxiliary maximum-flow formulation that identifies zero residual-capacity arcs and guides cut generation. Additional acceleration strategies, including cut strengthening, the ϵ-optimal solution approach, surrogate constraints, and a targeted arc generation procedure, are incorporated to improve computational performance. These enhancements improve scalability and enable the solution of large-scale instances that are otherwise difficult to solve using standard approaches.

Finally, the proposed model and solution methodology are evaluated through extensive computational experiments, including synthetic instances and a GIS-based case study of the Texas Gulf Coast. The results demonstrate the ability of the model to generate feasible and efficient distribution plans, evaluate infrastructure activation decisions jointly with routing choices, and identify capacity- and accessibility-sensitive regions under realistic disaster scenarios. These findings illustrate the practical applicability of the proposed framework for supporting coordinated strategic and tactical decision making in large-scale disaster relief operations.

3. Problem Definition

We propose a mixed-integer model that minimizes costs and satisfies total evacuees' demand by determining the relief supply locations and efficient delivery within a given time frame. Figure 1 illustrates the network design used in this study. Our network comprises three zones: distribution centers, transfer nodes, and shelters. Each node represents an area, such as a county, ZIP code, or census tract. In practice, transfer nodes are small towns along the route, highway junctions, gas stations, or county staging yards, and they capture shared access limits, queuing, and time windows. As shown in Figure 1, there are no arcs between distribution centers (i.e., no lateral shipments as all flow is directed toward shelters through the transfer nodes), but we assume that shelters can act as transfer nodes for other shelters as they also receive their supplies along a route.

Considering an underlying planar network (resembling a road network) with given locations of shelters and locations of potential distribution centers and transfer nodes that can be utilized, potential arcs and their capacities, shelters' demands, fixed costs of using arcs and nodes, flow-dependent variable costs at each node, transportation costs for each item, and supply capacities at distribution centers, our objective is to determine the optimal distribution centers to activate and paths composed of a sequence of transfer nodes from distribution centers (origins) to shelters (destinations) with the minimum cost to satisfy the worst case demand scenario for shelters, that is, when all shelters are open at maximum capacity. This worst case representation reflects common emergency logistics practice in which initial deployment plans are designed to remain feasible under peak shelter occupancy levels before demand uncertainty resolves during later operational stages (Dalal and Üster 2018).

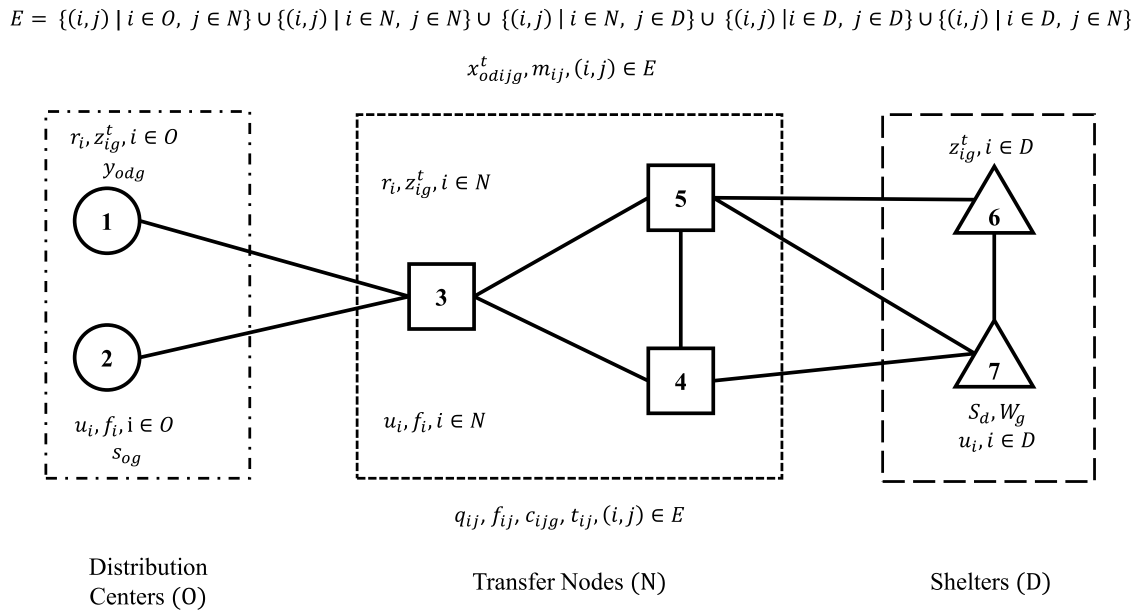
Capacities q_{ij} represent the maximum amount of relief flow that can traverse arc (i, j) within a single time period of the time-expanded network. Capacity is expressed in relief shipment units per period and, therefore, defines the maximum number of relief units that can be transported through the link during each period. Operationally, q_{ij} reflects physical and logistical throughput limits of the transportation infrastructure, such as lane availability, intersection throughput, access restrictions, or temporary reductions in usable road capacity during response operations.

Travel times t_{ij} are determined prior to optimization using road characteristics and expected operating conditions and remain fixed during optimization. Accordingly, q_{ij} functions solely as a throughput constraint that limits how much relief flow can traverse a link in each period. The model does not represent flow-dependent congestion effects in which routing decisions alter travel times. Instead, expected operational conditions are reflected through fixed travel-time inputs derived during preprocessing and through explicit capacity limits that represent restricted operational availability of roadway infrastructure during disaster response operations.

In our multiperiod planning, we consider a planning time horizon that roughly corresponds to the time frame from the beginning of the supply shipments to the end when all shelters receive their supplies. To model the transportation system's dynamics over that time frame, we discretize the planning time horizon into a set of periods. Then, to capture the spatiotemporal flow characteristics, we construct a time-expanded network that includes multiple copies of nodes and arcs from the original network for each period in the planning horizon. Assuming an average vehicle speed, we calculate estimated travel times for road segments based on distances between nodes and time units. Our goal is to satisfy the total demand within a specific time frame, which is the length of the planning horizon.

An illustrative time-expanded network with hypothetical flow is depicted in Figure 2, in which the rows list the locations from Figure 1, and columns are time periods. Three example flows with unit time arcs are shown for illustration. In the full model, when a link

Figure 1. Underlying Network Structure

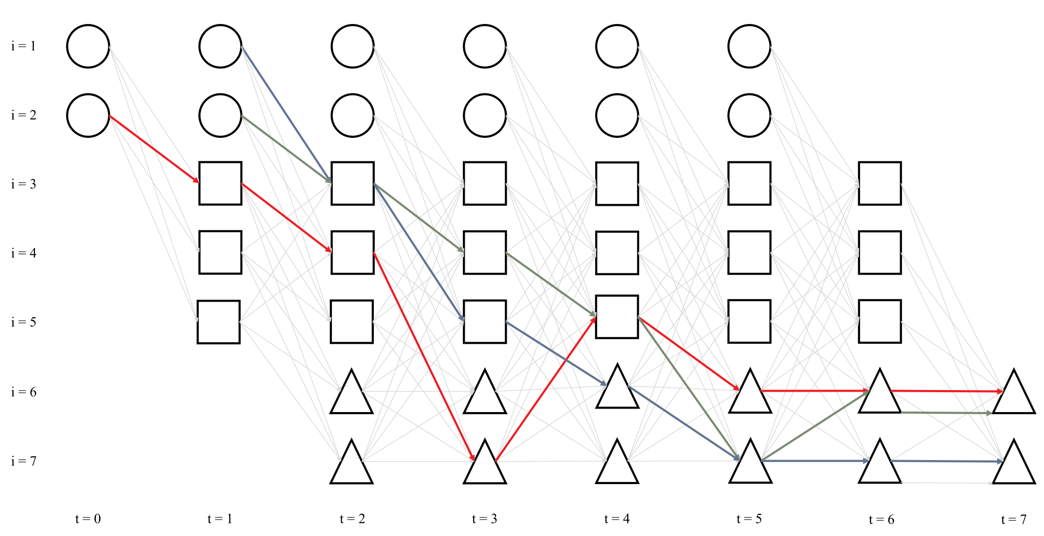


requires more than one period, the corresponding edge skips forward by that many columns.

We assume that there is no delay time at the nodes, meaning that reliefs are sent out from the transfer nodes in the same period that they are received, but similar to hub congestion models (Elhedhli and Hu 2005), we associate additional operational cost that represents cost of resources utilized to ensure uninterrupted flow of goods through a node (e.g., a town or city). This assumption treats transfer nodes as pass-through locations rather than storage facilities. In practice, intermediate locations are primarily used to redirect shipments without holding inventory, so intermediate storage and waiting are not explicitly modeled. Nodes with higher population density are assigned higher operational costs to reflect increased coordination effort, handling difficulty, and expected operational delays in densely populated areas.

Furthermore, relief supplies are modeled using standardized shipment units to provide a consistent measurement of flow across different item types. Multiple relief items are explicitly represented in the model; however, no priority structure is imposed among commodity types. This assumption reflects the initial relief phase, in which operational simplicity is emphasized and supplies are dispatched to meet minimum requirements across all item categories rather than dynamically prioritizing specific commodities. We assume that

Figure 2. (Color online) Dynamic Representation of the Study Problem



evacuees at shelters may have different demand levels for each item type; however, every demand unit (person, family, or group) requires at least a minimum quantity of each essential item. This assumption reflects common emergency logistics practice in which minimum relief requirements are defined per evacuee or household through standardized supply units to ensure baseline service coverage (Vanajakumari, Kumar, and Gupta 2016). To prevent undelivered items in the network, we assume that no new items should be sent from distribution centers after a specific time denoted as T_o within the planning horizon T. Operationally, this represents dispatch cutoff policies used in disaster response, in which outbound shipments are stopped after a certain time to ensure that deliveries arrive within the actionable response window and do not enter the network too late to be operationally useful within the planning horizon (Song, Chen, and Lei 2018).

One might consider modeling this as a simpler assignment problem, in which shelters are matched to the nearest DCs based on shortest paths. However, such an approach is insufficient because it ignores the critical role of network capacity and congestion. In a disaster, the shortest path may not be the best or even a feasible one if it becomes a bottleneck. Our three-tier network with decisions on which DCs to activate as the origins as well as the timing of flow for different products from these DCs through transfer nodes to shelters is explicitly designed to model these choke points and forces the model to select routes that are not only short but cost- and time-effective with sufficient capacity to handle the required flow of relief supplies. This distinction becomes particularly important when infrastructure activation and routing decisions are determined simultaneously because activating additional facilities or transportation links may relieve downstream capacity conflicts that cannot be resolved through routing decisions alone.

4. Model Formulation

We first introduce our notation in Table 1 for sets and their indices, input parameters, and the decision variables used in our mathematical model.

We assume that the shelter capacities are fully utilized in the worst case scenario, and thus, the total demand at each shelter d is estimated to equal $s_d w_g$. We also note that the parameter T_o represents the latest possible time for dispatching relief items from DCs. It is determined based on the minimum possible shortest path required to reach any shelter from any of the DCs, that is, $T_o = |\mathcal{T}| - \min_{d\in\mathcal{D}} \pi^*_d$. We estimate the travel time for every existing arc in the network by using road segment usage frequency and its capacity as well as the distance between the origin and destination. The number of times an arc $(i,j)\in\mathcal{A}$ is used in shortest paths (e.g., π_{od}) determines its road frequency. Higher frequency for a road indicates more traffic expected and, thus, longer travel times calculated, also factoring in the road segment capacity. We also scale real-valued travel times and round them to integers so that the time-expanded model is valid. Further details of travel time estimation are provided in Online Section S.2.1.

4.1. Model

Using the notation above and noting that, hereafter if not specified, $o\in\mathcal{O}$, $d\in\mathcal{D}$, $g\in\mathcal{G}$, $i\in\mathcal{I}$, $(i,j)\in\mathcal{A}$, and $\{i,j\}\in\mathcal{E}$, we formulate our problem **P** as follows:

$$P:\ \text{Min} \sum_{i\in\mathcal{I}\setminus\mathcal{D}} f_i r_i + \sum_{\{i,j\}\in\mathcal{E}} f_{ij} m_{ij} + \sum_{o\in\mathcal{O}}\sum_{d\in\mathcal{D}}\sum_{g\in\mathcal{G}}\sum_{(i,j)\in\mathcal{A}}\sum_{t\in\tau} c_{ijg} x^t_{odijg} + \sum_{g\in\mathcal{G}}\sum_{t\in\tau}\sum_{i\in\mathcal{I}} z^t_{ig} u_i \quad (1)$$

subject to

$$\sum_{d\in\mathcal{D}} y_{odg} \le 1 \qquad \forall o,g, \quad (2)$$

$$\sum_{d\in\mathcal{D}}\sum_{t=0}^{T_o-1}\sum_{j\in\mathcal{S}_o} x^t_{odojg} \le s_{og} r_o \qquad \forall o,g, \quad (3)$$

$$\sum_{o\in\mathcal{O}}\sum_{t=\pi^*_d}^{T-1}\left(\sum_{i\in\mathcal{P}_d} x^{t-t_{id}}_{odidg} - \sum_{i\in\mathcal{S}_d} x^t_{oddig}\right) \ge s_d w_g \qquad \forall d,g, \quad (4)$$

$$\sum_{t=0}^{T_o-1}\sum_{j\in\mathcal{S}_o} x^t_{odojg} = y_{odg} s_{og} \qquad \forall o,d,g, \quad (5)$$

$$\sum_{t=\pi_{od}}^{T-1}\left(\sum_{j\in\mathcal{S}_d} x^t_{oddjg} - \sum_{j\in\mathcal{P}_d} x^{t-t_{jd}}_{odjdg}\right) = -y_{odg} s_{og} \qquad \forall o,d,g, \quad (6)$$

$$\sum_{j\in\mathcal{S}_i} x^t_{odijg} - \sum_{j\in\mathcal{P}_i} x^{t-t_{ji}}_{odjig} = 0 \quad \forall o,d,g,i\in\mathcal{N}, t=\{\pi_{oi},\ldots,T-1\}, \quad (7)$$

$$\sum_{j\in\mathcal{P}_i, t\ge t_{ji}} x^{t-t_{ji}}_{odjig} - \sum_{j\in\mathcal{S}_i} x^t_{odijg} \ge 0 \quad \forall o,d,g,i\in\mathcal{D}, t=\{0,1,\ldots,T-1\}, \quad (8)$$

$$\sum_{o\in\mathcal{O}}\sum_{d\in\mathcal{D}}\sum_{g\in\mathcal{G}} (x^t_{odijg} + x^t_{odjig}) \le q_{ij} m_{ij} \quad \forall i<j, t=\{\pi^*_i,\ldots,T-1\}, \quad (9)$$

$$\sum_{o\in\mathcal{O}}\sum_{d\in\mathcal{D}}\sum_{j\in\mathcal{S}_i} x^t_{odijg} = z^t_{ig} \qquad \forall g,i,t=\{\pi^*_i,\ldots,T-1\}, \quad (10)$$

$$m_{ij} \le r_i \qquad \forall i\notin\mathcal{D}, j, \quad (11)$$

$$m_{ij} \le r_j \qquad \forall i, j\notin\mathcal{D}, \quad (12)$$

$$x^t_{odijg}, z^t_{ig} \ge 0 \qquad \forall o,d,g,(i,j),t, \quad (13)$$

$$0 \le y_{odg} \le 1 \qquad \forall o,d,g, \quad (14)$$

$$r_i, m_{ij} \in \{0,1\} \qquad \forall (i,j). \quad (15)$$

Objective Function (1) in model **P** minimizes the total cost, which includes fixed and variable costs. The first

Table 1. Notation

Sets and indices	
$\mathcal{O}$	Set of potential supply nodes (DCs), $o \in \mathcal{O}$
$\mathcal{N}$	Set of transfer nodes, $n \in \mathcal{N}$
$\mathcal{D}$	Set of destinations, $d \in \mathcal{D}$
$\mathcal{I}$	Set of all nodes, $i \in \mathcal{I}$
$\mathcal{T}$	Set of time periods, $t \in \mathcal{T}$
$\mathcal{E}$	Set of edges, $\{i,j\} \in \mathcal{E}$
$\mathcal{A}$	Set of directed arcs, $(i,j) \in \mathcal{A}$
$\mathcal{P}_i$	Set of node i's predecessor nodes, $j \in \mathcal{P}_i$
$\mathcal{S}_i$	Set of node i's successor nodes, $j \in \mathcal{S}_i$
$\mathcal{G}$	Set of items, $g \in \mathcal{G}$
τ_o	Set of allowable times for leaving source node $o \in \mathcal{O}$, $t \in \tau_o = \{0, 1, \ldots T_o\}$
Parameters	
s_{og}	Capacity of supply node o for item g, $o \in \mathcal{O}$, $g \in \mathcal{G}$
s_d	Capacity of shelter, $d \in \mathcal{D}$
w_g	Demand of relief item g per person, $g \in \mathcal{G}$
q_{ij}	Maximum relief flow that can traverse arc (i,j) within one time period, $(i,j) \in \mathcal{A}$
t_{ij}	Travel time along arc, $(i,j) \in \mathcal{A}$
f_i	Fixed cost for including nodes (origins and transfers), $i \in \mathcal{I} \backslash \mathcal{D}$
u_i	Variable cost at i for each unit of flow, $i \in \mathcal{I}$
f_{ij}	Fixed cost for including edge, $\{i,j\} \in \mathcal{E}$
c_{ijg}	Variable cost for unit flow of $g \in \mathcal{G}$ on arc, $(i,j) \in \mathcal{A}$
π_{oi}	Shortest path length from a supply origin o to node i, $o \in \mathcal{O}$, $i \in \mathcal{I}$
π_i^*	Shortest path length to node i from an origin $o(\pi_i^* = \min_{o \in \mathcal{O}}\{\pi_{oi}\})$, $i \in \mathcal{I}$
$freq_{ij}$	Number of times the arc (i,j) are utilized in SP from DCs to shelters, $(i,j) \in \mathcal{A}$
Decision variables	
r_i	One if node i is opened, zero otherwise, $i \in \mathcal{I} \backslash \mathcal{D}$
m_{ij}	One if edge $\{i,j\}$ is used, zero otherwise, $\{i,j\} \in \mathcal{E}$
z_{ig}^t	Total amount of item g forwarded at node i at time t, $g \in \mathcal{G}$, $i \in \mathcal{I}$, $t \in \tau$
x_{odijg}^t	Amount of item g flow from o to d on arc (i,j) at time t, $g \in \mathcal{G}$, $o \in \mathcal{O}$, $d \in \mathcal{D}$, $(i,j) \in \mathcal{A}$, $t \in \tau$
y_{odg}	Proportion of item g from node o is sent to node d, $g \in \mathcal{G}$, $o \in \mathcal{O}$, $d \in \mathcal{D}$

and second terms represent the fixed cost of using nodes (DCs and transfer nodes) and arcs in the final network. The last two terms give the total variable cost on arcs and nodes.

Constraints (2) restrict the allocation of items to each DC so that the available DC capacity is not violated. Constraints (3) ensure that the total outbound flow from each activated DC within the permissible time frame does not surpass its capacity for each item. The outgoing flow capacity of a nonoperational DC is zero for all periods. Constraint (4) ensures that the needs of evacuees at a shelter are fulfilled. As depicted in Figure 2, shelters may also function as transfer nodes for other nodes. Hence, the net flow difference at each shelter across periods reflects the actual relief received by the shelter. It is important to note that the initial flow cannot reach shelter d before time π_d^* as implied by the second sum in (4). Constraints (5) and (6) dictate the aggregate flow of each item departing from the DCs within the allowed period and the total amount received by shelters at the end of the designated time window, respectively. Constraints (7) are the flow balance at pure transfer nodes at each period, stipulating that transfer nodes neither store nor delay any items; they must forward items concurrently with their reception. Constraints (8) are for flow balance at the shelters (destinations that can also act as transfer nodes). They ensure that no shelter node can send relief items at any time unless it receives that flow simultaneously. This prevents shelters from independently producing and distributing items. Constraints (9) specify that the flows on each arc must not exceed their capacities. If m_{ij} is zero, then the capacity of arc (i,j) is zero, showing that this arc is excluded from the final solution. Conversely, if m_{ij} is one, indicating that the arc is active, then the total inflow to node i and outflow from node i must be within the assigned arc capacity at time t. Constraint (10) defines the items loaded at each node at any time t as it is required to determine the associated cost for each node that transfers flow. Constraints (11) and (12) ensure that if an arc is active, its origin and destination nodes must also be active. Lastly, Constraints (13)–(15) state the variable range requirements.

5. Solution Methodology

Because our problem is modeled on a time-expanded network, large-scale instances pose an additional challenge for its solution. To motivate an efficient approach based on BD (Benders 1962), we first observe that, when the values of binary variables r_i and m_{ij} are fixed, that is, transfer node locations and arcs utilized specified, respectively, our model **P** reduces to a linear programming model with only continuous variables. However, as mentioned above, we observe that infeasibility in subproblems, mainly because of an unconnected underlying network and insufficient arc capacities in a solution provided by the master problem (i.e., fixed r_i and m_{ij} values) significantly increases total runtimes. Therefore, we devise and incorporate several enhancement techniques for our problem and its solution via an efficient BD framework. Next, we describe our general framework with its enhancements. We note that some of these enhancements can also be applicable in other network design contexts, especially on time-expanded networks.

5.1. Benders Decomposition Framework

BD is an iterative algorithm for mixed-integer programs that works by decomposing them into a master problem **MP** and a subproblem **SP**. Typically, the **MP** includes all integer variables, whereas the **SP** handles the continuous variables, including one that connects the two problems. The **SP** is solved using the solution from the **MP**, and optimality or feasibility cuts are added based on its dual solution to the **MP**. Solving the **MP** and **SP** generates lower and upper bounds (for a min problem), respectively, and the process continues until the optimality gap is small or a threshold runtime is reached. However, because of the slow convergence of regular BD, it is necessary to improve the efficiency of this algorithm by employing acceleration approaches as described in Section 5.2.

5.1.1. Subproblem and Its Dual. By decomposing our problem in a way that the location and arc selection variable values, that is, for r_i and m_{ij} variables, are determined in the **MP** to be $\hat{r}_i$, $\hat{m}_{ij}$, respectively, we obtain the subproblem **SP** as follows:

$$SP: \text{ Min} \sum_{o\in\mathcal{O}}\sum_{d\in\mathcal{D}}\sum_{g\in\mathcal{G}}\sum_{(i,j)\in\mathcal{A}}\sum_{t\in\tau} c_{ijg}x^t_{odijg} + \sum_{g\in\mathcal{G}}\sum_{t\in\tau}\sum_{i\in\mathcal{I}} z^t_{ig}u_i \tag{16}$$

subject to

$$\sum_{d\in\mathcal{D}} y_{odg} \le 1 \qquad \forall o,g, \tag{17}$$

$$\sum_{d\in\mathcal{D}}\sum_{t=0}^{T_o-1}\sum_{j\in\mathcal{S}_o} x^t_{odojg} \le s_{og}\hat{r}_o \qquad \forall o,g, \tag{18}$$

$$\sum_{o\in\mathcal{O}}\sum_{t=\pi^*_d}^{T-1}\left(\sum_{i\in\mathcal{P}_d} x^{t-t_{id}}_{odidg} - \sum_{i\in\mathcal{S}_d} x^t_{oddig}\right) \ge s_d w_g \qquad \forall d,g, \tag{19}$$

$$\sum_{t=0}^{T_o-1}\sum_{j\in\mathcal{S}_o} x^t_{odojg} = y_{odg}s_{og} \qquad \forall o,d,g, \tag{20}$$

$$\sum_{t=\pi_{od}}^{T-1}\left(\sum_{j\in\mathcal{S}_d} x^t_{oddjg} - \sum_{j\in\mathcal{P}_d} x^{t-t_{jd}}_{odjdg}\right) = -y_{odg}s_{og} \qquad \forall o,d,g, \tag{21}$$

$$\sum_{j\in\mathcal{S}_i} x^t_{odijg} - \sum_{j\in\mathcal{P}_i} x^{t-t_{ji}}_{odjig} = 0 \qquad \forall o,d,g,i\in\mathcal{N},\ t=\{\pi_{oi},\ldots,T-1\}, \tag{22}$$

$$\sum_{j\in\mathcal{P}_i, t\ge t_{ji}} x^{t-t_{ji}}_{odjig} - \sum_{j\in\mathcal{S}_i} x^t_{odijg} \ge 0 \qquad \forall o,d,g,i\in\mathcal{D},\ t=\{0,1,\ldots,T-1\}, \tag{23}$$

$$\sum_{o\in\mathcal{O}}\sum_{d\in\mathcal{D}}\sum_{g\in\mathcal{G}}(x^t_{odijg} + x^t_{odjig}) \le q_{ij}\hat{m}_{ij} \qquad \forall i<j,\ t=\{\pi^*_i,\ldots,T-1\}, \tag{24}$$

$$\sum_{o\in\mathcal{O}}\sum_{d\in\mathcal{D}}\sum_{j\in\mathcal{S}_i} x^t_{odijg} = z^t_{ig} \qquad \forall g,i,t=\{\pi^*_i,\ldots,T-1\}, \tag{25}$$

$$x^t_{odijg}, z^t_{ig} \ge 0 \qquad \forall o,d,g,(i,j),t, \tag{26}$$

$$0 \le y_{odg} \le 1 \qquad \forall o,d,g. \tag{27}$$

With given $\hat{r}_i$ and $\hat{m}_{ij}$ values and defining dual variables associated with Constraints (17)–(25) as μ_{og}, λ_{og}, α_{dg}, β_{odg}, ψ_{odg}, γ^t_{odig}, η^t_{odig}, θ^t_{ij}, and Δ^t_{ig}, respectively, we obtain the dual subproblem, **DSP** as given in the Online Section S.1. Letting $\mathcal{V}$ and $\mathcal{U}$ denote all extreme points and extreme rays of the **DSP** polyhedron and $\hat{\mu}^v_{og}$, $\hat{\lambda}^v_{og}$, $\hat{\alpha}^v_{dg}$, $\hat{\beta}^v_{odg}$, $\hat{\psi}^v_{odg}$, $\hat{\gamma}^{tv}_{odig}$, $\hat{\eta}^{tv}_{odig}$, $\hat{\theta}^{tv}_{ij}$, and $\hat{\Delta}^{tv}_{ig}$ be the associated dual values with D^v as the objective value for $v\in\mathcal{V}$ when $\hat{\mu}^u_{og}$, $\hat{\lambda}^u_{og}$, $\hat{\alpha}^u_{dg}$, $\hat{\beta}^u_{odg}$, $\hat{\psi}^u_{odg}$, $\hat{\gamma}^{tu}_{odig}$, $\hat{\eta}^{tu}_{odig}$, $\hat{\theta}^{tu}_{ij}$, and $\hat{\Delta}^{tu}_{ig}$ are the dual values for $u\in\mathcal{U}$, we obtain the Benders cuts as follows. If **DSP** is bounded (i.e., **SP** is feasible), then **DSP** can be written as $\min_{D\ge 0}\{D : D \ge D^v,\ \forall v\in\mathcal{V}\}$, and the Benders optimality cut is stated as follows:

$$D \ge -\sum_{o\in\mathcal{O}}\sum_{g\in\mathcal{G}}(\hat{\mu}^v_{og} + s_{og}r_o\hat{\lambda}^v_{og}) + \sum_{d\in\mathcal{D}}\sum_{g\in\mathcal{G}} s_d w_g \hat{\alpha}^v_{dg} - \sum_{o\in\mathcal{O}}\sum_{(i,j)\in\mathcal{A}}\sum_{t\in\tau} q_{ij}m_{ij}\hat{\theta}^{tv}_{ij}\ v\in\mathcal{V}. \tag{28}$$

If **DSP** is unbounded (i.e., **SP** is infeasible), then the Benders feasibility cut is obtained as

$$0 \ge -\sum_{o\in\mathcal{O}}\sum_{g\in\mathcal{G}}(\hat{\mu}^u_{og} + s_{og}r_o\hat{\lambda}^u_{og}) + \sum_{d\in\mathcal{D}}\sum_{g\in\mathcal{G}} s_d w_g \hat{\alpha}^u_{dg} - \sum_{o\in\mathcal{O}}\sum_{(i,j)\in\mathcal{A}}\sum_{t\in\tau} q_{ij}m_{ij}\hat{\theta}^{tu}_{ij}\ u\in\mathcal{U}. \tag{29}$$

5.1.2. Reformulation and the Master Problem. With the above representation of **DSP** using the extreme points and extreme rays of its polyhedron, we reformulate the overall problem **P** as

$$\begin{aligned}\text{Min} \sum_{i\in\mathcal{I}\setminus\mathcal{D}} f_i r_i + \sum_{\{i,j\}\in\mathcal{E}} f_{ij}m_{ij} + D \\ \text{s.t. } (11),(12),(15),(28),(29).\end{aligned} \tag{30}$$

Because Constraints (28) and (29) are not readily available, in the BD framework, they are relaxed, and the appropriate one is iteratively added whenever a **DSP** is solved for a fixed **MP** solution. The problem **MP** is the relaxed overall reformulation, and Constraints (28) and (29) are generated in a delayed fashion. Because **MP** is a relaxation of **P**, it provides a lower bound to **P**. On the other hand, the fixed **MP** solution and the associated **SP** solution together prescribe a feasible solution and, thus, an upper bound for **P**. Building on this BD framework, our overall algorithm is provided after we discuss our acceleration techniques in detail in Section 5.2.

5.1.3. An Alternative Formulation for SP Separability. One of BD's important features that makes it a preferable solution algorithm for large-scale problems is its ability to decompose one large problem into many small problems. Being able to separate subproblem(s) into even smaller and identically formed problems helps solve the original problem with faster solution times. By closely examining our original problem **P**, one can see that all constraints that become a part of **SP** are written for each relief item $g \in \mathcal{G}$ except for Constraint (9) (Constraint (24) in **SP**), which is the road capacity constraint. Therefore, by defining an auxiliary variable, σ^t_{ijg}, as the upper limit for road capacity assigned to each item at a specific time, we can replace this constraint in **P** by the following constraints:

$$\sum_{o\in\mathcal{O}}\sum_{d\in\mathcal{D}}(x^t_{odijg}+x^t_{odjig}) \leq \sigma^t_{ijg} \;\; \forall i<j, t=\{\pi^*_i,\ldots,T-1\}, g, \tag{31}$$

$$\sum_{g\in\mathcal{G}}\sigma^t_{ijg} = q_{ij}m_{ij} \;\; \forall i<j, t=\{\pi^*_i,\ldots,T-1\}. \tag{32}$$

Constraint (31) applies upper limits for each item $g \in \mathcal{G}$, and Constraint (32) ensures that these upper limits do not exceed the road capacity of their corresponding arcs in the network. We can consider $\sigma^t_{ijg} \geq 0$ as decision variables in **MP** and dispatch Constraints (31) and (32) to **SP** and **MP**, respectively. The new **SP** now becomes separable for each relief item $g \in \mathcal{G}$, lending itself to multiple cuts, one for each relief item g, at each iteration of the BD algorithm.

Formulation of the dual subproblem along with the optimality and feasibility cuts changes accordingly. Letting D^v_g be the objective function of each subproblem g, $\mathbf{DSP}_\mathrm{g}$ can be written as $\min_{D_g\geq 0}\{D_g : D_g \geq D^v_g, \; \forall v \in \mathcal{V}, g \in \mathcal{G}\}$ as given explicitly in the Online Section S.1. Then, new Benders optimality cuts,

$$\begin{aligned} D_g \geq &-\sum_{o\in\mathcal{O}}(\hat{\mu}^v_{og}+s_{og}r_o\hat{\lambda}^v_{og}) + \sum_{d\in\mathcal{D}} s_d w_g \hat{\alpha}^v_{dg} \\ &-\sum_{o\in\mathcal{O}}\sum_{(i,j)\in\mathcal{A}}\sum_{t\in\tau}\sigma^t_{ijg}\hat{\theta}^{tv}_{ij} \qquad v\in\mathcal{V}\, g\in\mathcal{G}, \end{aligned} \tag{33}$$

and feasibility cuts,

$$\begin{aligned} 0 \geq &-\sum_{o\in\mathcal{O}}(\hat{\mu}^u_{og}+s_{og}r_o\hat{\lambda}^u_{og}) + \sum_{d\in\mathcal{D}} s_d w_g \hat{\alpha}^u_{dg} \\ &-\sum_{o\in\mathcal{O}}\sum_{(i,j)\in\mathcal{A}}\sum_{t\in\tau}\sigma^t_{ijg}\hat{\theta}^{tu}_{ij} \qquad u\in\mathcal{U}, g\in\mathcal{G}, \end{aligned} \tag{34}$$

are obtained. The reformulation of **MP** is stated as

$$\begin{aligned} &\text{MP: Min} \sum_{i\in\mathcal{I}\setminus\mathcal{D}} f_i r_i + \sum_{\{i,j\}\in\mathcal{E}} f_{ij}m_{ij} + \sum_{g\in\mathcal{G}} D_g \\ &\text{subject to (11),(12),(15),(32),(33),(34)}, \; \sigma^t_{ijg} \geq 0 \\ &\qquad \forall i<j, t=\{\pi^*_i,\ldots,T-1\}, g\in\mathcal{G}. \end{aligned} \tag{35}$$

5.2. Acceleration Methods

To solve our model, we initially experiment with the default automated BD algorithm, provided by the CPLEX callable library, which employs some acceleration methods including lazy constraint callbacks with parallel implementation, normalization, stabilization, etc. (Bonami, Salvagnin, and Tramontani 2020). We observe in the performance results that there is a need to enhance its algorithmic efficiency regarding solution quality and time. This is primarily because the automated BD is designed to solve problems with generic improvement methods and lacks specific problem-related information. The advantage of employing the BD framework usually presents itself as the opportunity to integrate it with other approaches to increase efficiency. This is generally achieved by addressing the difficulties in obtaining reasonable bounds via addressing the separate problems, **MP** and **SP**, that provide the bounds for the problem at hand. Thus, in this section, we present approaches to incorporate into the BD framework to enhance its performance for our problem, noting that some enhancements apply in other network design settings, summarized as follows:

- In Section 5.2.1, we devise an approach that is based on the capacity-relaxed version of **P**, a maximum flow problem, and its corresponding zero residual arcs as well as a reduced network design problem to modify the underlying network provided by the **MP** so that it allows a feasible solution for the **SP** and Benders optimality cuts.
- In Section 5.2.2, we apply the ϵ-optimal method in solving **MP** to only feasibility so that its solution time is reduced.
- In Section 5.2.3, we observe an alternate optimal solution for each subproblem, each of which can generate a different optimality cut. Therefore, we strengthen the optimality cuts obtained via $\mathbf{DSP}_\mathrm{g}$ so that the **MP** provides improved lower bounds for faster convergence. Because **MP** is an integer model, solving it is usually the most time-consuming part of the overall approach.

- In Section 5.2.4, we devise and incorporate surrogate constraints into our **MP** to assist it further in generating solutions that potentially lead to feasible primal subproblems, thereby minimizing the need for generating feasibility cuts.

5.2.1. Subproblem Feasibility Check and Optimality Cuts. As mentioned earlier, dealing with the infeasibility of subproblems by generating feasibility cuts is inefficient; therefore, we seek **MP** solutions that avoid the infeasibility of primal subproblems. In our initial computational experiments, we notice that a disconnected network and inadequate arc capacities provided by the **MP** solution are the main reasons for infeasible subproblems. In the first iteration of BD, solving an arc capacity–relaxed version of the original problem initially and fixing its r_i and m_{ij} variable values helps to have a connected underlying network for solving the subproblem. Despite addressing connectivity, we note that this still may not provide enough capacity on the utilized arcs given by m_{ij}. This latter infeasibility is revealed once the subproblem is solved. However, our subproblem itself is large and requires significant computational effort for its solution; thus, it is inefficient to solve the **SP** for this purpose. Thus, before solving the subproblem, we consider checking the feasibility of the subproblem efficiently on the underlying network provided by the solution of the master problem. Suppose we observe **SP** infeasibility because of insufficient arc capacities. In that case, we need to modify the **MP** solution before solving **DSP** to avoid relying on feasibility cuts.

For checking the feasibility of the overall primal **SP** without fully solving it, we devise and solve a series of maximum flow problems, one for each item g. To do so, for each item g, we define a super supply node and a super demand node, connecting the super supply node to the DCs and the shelters to the super demand node. We set the capacities of the arcs connecting the super supply node to each DC to the corresponding supply capacities for that item at the DCs. For the demand side, we consider the shelters' demands for the item as the capacities of the arcs connecting them to the super demand node. Notice that, whereas the capacities on these added arcs are item-specific, the capacities on the arcs of the original network are not.

Furthermore, considering the time-expanded nature of our original problem network, it is crucial to capture the effect of the time dimension on arc capacities and the residual graph to reduce the model complexity for its quick solution. To achieve this, we first identify the widest possible time window for an arc (i,j) to be utilized. Let $a(i)$ be the shortest time to reach node i from the DCs and $b(i)$ be the shortest time to reach the closest shelter from node i and recall that the time horizon is T. Then, an arc (i,j) can be utilized in a time window $[a(i), T-b(j)]$, thereby simplifying the temporal aspects of the model for our purposes and leading to a calculation of time-accounted arc capacity, Cap_{ij}, as

$$Cap_{ij} = q_{ij} \times [T - a(i) - b(j)] \qquad \forall (i,j). \tag{36}$$

This equation ensures that the arc capacities reflect the network's physical flow limits and time constraints. Integrating the time dimension into the capacity calculations ensures that resources are allocated effectively within the network's temporal limitations, reducing the model complexity.

To check **SP** feasibility, we apply the augmenting path max-flow algorithm (Ford–Fulkerson method) for each item g on the smaller network with super supply and demand nodes, utilizing the time-accounted capacities to find the maximum feasible flow. After processing an item, we update the residual network to reflect the capacity levels accurately in subsequent iterations. Then, continuing with the following item, we define new super supply and demand nodes and their associated arcs connecting the original residual network and apply the augmenting path algorithm again. After determining the maximum flow for an item, we check the feasibility of the subproblem for that item by examining the residual capacities on the arcs connecting the shelters to the super demand node. Recall that the capacity on such an arc equals the demand for the item at its corresponding shelter. If all these residual capacities are zero, it implies that the demands for the item across all shelters can be met using the solution obtained from the master problem. This also indicates that the minimum cut for the network associated with the current item consists of the arcs connecting shelters to the super demand node. Conversely, if any of these arcs have nonzero residual capacities, then the subproblem is infeasible, and thus, it is necessary to adjust the master problem solution. This latter situation indicates that one or more arcs in the original network have zero residual capacity. These critical arcs, which have zero capacity in the residual network, restrict the maximum flow that saturates super demand arcs (i.e., satisfy the demand for the item at the shelters). Therefore, to reach **SP** feasibility, we introduce new arcs into the underlying network provided by the **MP** solution.

To increase maximum flow for each item, we consider additional (candidate) arcs that are not in the **MP** solution but provide new paths to the demand (shelter) node for which the item's demand is not satisfied. We prefer to select the most cost-effective entries among candidates, satisfying the item's demand and minimizing the total cost. To do so, considering the collectively exhaustive, mutually exclusive three arc subsets in the original network, ($\mathcal{E}_E \cup \mathcal{E}_C \cup \mathcal{E}_N = \mathcal{E}$), we solve a reduced-size fixed-charge network design problem for the item with modified arc costs as follows:

Arcs provided by the **MP** solution ($\mathcal{E}_E = \{(i,j)|\hat{m}_{ij} = 1\}$): We set the cost of these arcs to zero because we want to ensure that these arcs and nodes are prioritized for use. The capacities of these arcs are the same as Cap_{ij} (36).

Candidate arcs ($\mathcal{E}_C = \{(i,j)|\hat{m}_{ij} = 0, i\,\textit{reachable}, j$ not *reachable in residual graph*$\}$): The fixed and variable costs of these arcs are as given in the original network. The capacities of these arcs are also calculated by (36).

Noncandidate arcs ($\mathcal{E}_N = \mathcal{E} \setminus \{\mathcal{E}_E \cup \mathcal{E}_C\}$): These arcs must not be selected. Thus, their corresponding costs are set to infinity with zero capacity.

Letting h_{ij} and c_{ij} denote the fixed and variable costs as set above for the arc subsets in the network, respectively; x_{ij} be the flow on arc (i,j); y_{ij} be a binary variable that takes the value one if arc (i,j) is selected; and u_{ij} be the capacity of arc (i,j), we solve the following problem for the item g to adjust the **MP** solution by adding arcs for increased network:

$$\text{Min} \sum_{(i,j)\in\mathcal{E}} h_{ij}y_{ij} + c_{ij}x_{ij} \tag{37}$$

subject to

$$x_{ij} \le u_{ij}y_{ij} \qquad \forall (i,j) \in \mathcal{E}, \tag{38}$$

$$\sum_{j\in\mathcal{S}_o} x_{oj} \le r_o S_{og} \qquad \forall o, \tag{39}$$

$$\sum_{j\in\mathcal{P}_d} x_{jd} - \sum_{j\in\mathcal{S}_d} x_{dj} \ge S_d W_g \qquad \forall d, \tag{40}$$

$$\sum_{j\in\mathcal{P}_i} x_{ji} - \sum_{j\in\mathcal{S}_i} x_{ij} = 0 \qquad \forall i \in \mathcal{N}, \tag{41}$$

$$y_{ij} \in \{0,1\} \qquad \forall (i,j), \tag{42}$$

$$x_{ij} \ge 0 \qquad \forall (i,j). \tag{43}$$

Objective Function (37) minimizes the fixed cost of using arcs and the variable cost of flow. Notice that several of the variables in this formulation have preset values, specifically, $y_{ij} = 1$, $\forall (i,j) \in \mathcal{E}_E$ and $y_{ij} = 0$, $\forall (i,j) \in \mathcal{E}_N$. Constraints (38) and (39) satisfy arc and supply capacity limitations, respectively. Constraints (40) ensure that all shelter demands are fulfilled. Constraints (41) are flow balance constraints at transfer nodes. Constraints (42) and (43) enforce binary and nonnegativity requirements.

Once we obtain the additional arcs, we update the residual graph to solve (37)–(43) to obtain the new flows for the item at hand. We then incorporate these additionally selected arcs with updated capacities into the underlying network and start the same process for the following item.

We remark that (i) because our objective is to identify an underlying network for the **SP** efficiently, rather than optimizing flows, the order in which the items are considered does not matter in our case, and furthermore, (ii) although our approach for this purpose does not guarantee feasibility of the overall subproblem at each BD iteration, it significantly reduces the number of infeasible solutions to avoid Benders feasibility cuts and, thus, provides a computational advantage.

We summarize the overall approach of generating feasible solutions for subproblems in Algorithm 1.

Algorithm 1 (Feasibility Check and Arc Generation)

Initialize initial underlying network, r_i and m_{ij} values, from **RMP**
Let $\mathcal{O}_s$ as selected DCs and $\mathcal{N}_s$ be selected transfer nodes by **RMP**
Define SP_{ij} as the shortest path from i to j, where $i,j \in \mathcal{I}$.
Calculate $a(o) = 0$, $b(o) = \min_{d\in\mathcal{D}} SP_{od}$, $\quad \forall o \in \mathcal{O}_s$.
Calculate $a(i) = \min_{o\in\mathcal{O}_s} SP_{oi}$, $b(i) = \min_{d\in\mathcal{D}} SP_{id}$, $\forall i \in \mathcal{N}_s$.
Calculate $a(d) = \min_{o\in\mathcal{O}_s} SP_{od}$, $b(d) = 0$, $\quad \forall d \in \mathcal{D}$.
Calculate $Cap_{ij} = q_{ij}[T - b(j) - a(i)] \quad \forall \{i,j\} \in \mathcal{O}_s \times \mathcal{N}_s, \mathcal{N}_s \times \mathcal{N}_s, \mathcal{N}_s \times \mathcal{D}, \mathcal{D} \times \mathcal{D}$
for each item g **do**
  Define and connect super supply (p) and demand (q) nodes to DCs and shelters, respectively.
  Calculate $Cap_{po} = s_{og} \quad \forall o \in \mathcal{O}_s$ and $Cap_{dq} = s_d w_g \quad \forall d \in \mathcal{D}$.
  Apply the augmented path algorithm on the expanded network to find a max flow from p to q.
  Update residual capacities as $Cap_{ij} = -Flow_{ij}$, where $Flow_{ij}$ is obtained flow on arc (i,j).
  if $\exists (d,q)$ such that $Cap_{dq} > 0$, **then**
    for $(i,j) \in \mathcal{E}_E$ **do**
      $h_{ij} = 0$, $c_{ij} = 0$, $u_{ij} = Cap_{ij}$.
    end for
    for $(i,j) \in \mathcal{E}_C$ **do**
      $h_{ij} = f_{ij}$, $c_{ij} = c_{ijg}$, $u_{ij} = q_{ij}[T - b(j) - a(i)]$.
    end for
    for $(i,j) \in \mathcal{E}_N$ **do**
      $h_{ij} = \infty$, $c_{ij} = \infty$, $u_{ij} = 0$.
    end for
    Solve reduced network design, (37)–(43), with updated costs and capacities
    Update $\hat{m}_{ij} = 1$ if $y_{ij} = 1$.
  end if
end for
Return the most recent $\hat{m}_{ij}$ and $\hat{r}_i$.

5.2.2. ϵ-Optimal Method. Whereas the subproblem separation helps with the convergence of the overall algorithm, it also causes an increase in both complexity and runtime of the **MP**, especially in later iterations. To address this issue, we integrate the ϵ-optimal method to solve the **MP** with an additional constraint to only feasibility rather than optimality (Geoffrion and Graves 1974). Specifically, Constraint (45), whose left-hand side is the **MP** objective function, is included in the **MP** formulation, and this provides a lower bound for our problem. Because **MP** is now only a solving-to-feasibility problem, we nullify the objective function and solve the following reformulated **MP** model:

$$\textbf{RMP}: \text{ Min } 0 \tag{44}$$

subject to

$$\sum_{i\in\mathcal{I}\setminus\mathcal{D}} f_i r_i + \sum_{\{i,j\}\in\mathcal{E}} f_{ij} m_{ij} + \sum_{g\in\mathcal{G}} D_g \le UB(1-\epsilon)$$

$$(11), (12), (15), (32), (33), (34),\ \sigma^t_{ijg} \ge 0$$

$$\forall i < j, t = \{\pi^*_i, \ldots, T-1\}, g \in \mathcal{G}, \tag{45}$$

where the parameter ϵ denotes the desired optimality gap for the overall model and UB is the current best upper bound. Then, at each iteration, instead of solving the master problem to optimality, we only seek a feasible solution with an objective function less than or equal to $UB(1-\epsilon)$. The iterations are continued until **RMP** cannot provide a feasible solution, which can only be because of (45). Having iterations completed faster this way, the overall runtime decreases significantly with a guarantee that the optimality gap is within 100ϵ.

5.2.3. Cut Strengthening. In our numerical experiments, we observe that, because of the degeneracy of the study problem, subproblems at each iteration have multiple optimal solutions, each of which can generate distinct optimality cuts. Therefore, it is critical to identify the stronger optimality cuts to help with faster convergence. To strengthen our Bender's optimality cuts, we proceed similarly to the two-phase approach utilized in Van Roy (1986) and Üster and Agrahari (2011).

Recall that we utilize a BD with cut disaggregation in which we have one cut for each g item, (33) or (34), obtained by solving $\mathbf{DSP}_g$. For an optimization problem $\min_{y\in Y, z\in\mathbb{R}}\{f(w) + yg(w),\ \forall w \in W\}$, a cut $z \ge f(v) + yg(v)$ is stronger than another cut $z \ge f(w) + yg(w)$ if $f(v) + yg(v) \ge f(w) + yg(w),\ \forall y \in Y$, and for at least one $y \in Y$, we have $f(v) + yg(v) > f(w) + yg(w)$ (Magnanti and Wong 1981). Based on this, we solve $\mathbf{DSP}_g$ in two phases. In phase I, with $\hat{m}_{ij}$, $\hat{r}_i$, and $\hat{\sigma}^t_{ijg}$ obtained from the **MP** solution, we solve a $\mathbf{DSP}_g$ for each item and obtain dual variables μ_{og}, λ_{og}, α_{dg}, β_{odg}, ψ_{odg}, γ^t_{odig}, η^t_{odig}, θ^t_{ij}, and Δ^t_{ig}. We compute the upper bound of our problem using the phase I solution in the usual manner. It is clear that dual variables associated with zero coefficients in the dual objective function (i.e., zero binary solution values from **MP**) have no impact on the optimal objective value as long as they are feasible. Therefore, to ensure feasibility of the optimal solution, we consider all dual variables in phase I.

In phase II, we fixed the dual values corresponding to the terms appearing in the dual objective function with nonzero coefficients and considered the remaining ones as the decision variables. Once the optimal solution of phase I is fixed, dual variables associated with zero coefficient have no impact on the optimal solution; therefore, the phase II problem can determine their values in such a way that the Benders cut generated is stronger in the sense of Magnanti and Wong (1981). After solving these two phases, we constructed BD optimality cuts for each item with the help of the final optimal values of the dual variables and added them to **MP**.

5.2.4. Surrogate Constraints. We observe that solving the **MP** by relaxing the arc capacity constraints initially for the first iteration (as mentioned in Section 5.2.1) and employing the ϵ-optimal approach (Section 5.2.2) may still generate bounds that lead to slow convergence. We introduce surrogate constraints to enhance the lower bounds and incorporate them into the master problem. These surrogate constraints are redundant for the original problem but provide valuable information about the subproblems.

5.2.4.1. Partial Objective–Based Surrogate Constraints. We observe that our initial model **P** is a combination of three problems: node selection, arc selection, and network flow. We define these three problems based on their objective functions, respectively, as follows:

$$RP1: \quad \text{Min} \sum_{i\in\mathcal{I}\setminus\mathcal{D}} f_i r_i \qquad s.t: (2)-(15)$$

$$RP2: \quad \text{Min} \sum_{\{i,j\}\in\mathcal{E}} f_{ij} m_{ij} \qquad s.t: (2)-(15)$$

$$RP3: \quad \text{Min} \sum_{o\in\mathcal{O}} \sum_{d\in\mathcal{D}} \sum_{g\in\mathcal{G}} \sum_{(i,j)\in\mathcal{A}} \sum_{t\in\tau} c_{ijg} x^t_{odijg} + \sum_{g\in\mathcal{G}} \sum_{t\in\tau} \sum_{i\in\mathcal{I}} z^t_{ig} u_i \qquad s.t: (2)-(15).$$

The problem **RP3** contains decision variables from subproblems, and **RP2** is much harder to solve than **RP1**. Therefore, we only use **RP1** to generate a surrogate constraint, which is valid for the original problem **P**, given as

$$\sum_{i\in\mathcal{I}\setminus\mathcal{D}} f_i r_i \ge W^{PA}, \tag{46}$$

where W^{PA} is the optimal solution of **RP1**. The validity of this constraint can be argued following the proof given in Lin and Üster (2014).

5.2.4.2. LP Relaxation. As with the above surrogate constraints, we can use the optimal solution of problems with combined partial objective functions as surrogate constraints in the master problem **P**. Considering the original problem with respect to all fixed costs as the objective function, we have

$$\mathbf{RP4}: \quad \text{Min} \sum_{i\in\mathcal{I}\setminus\mathcal{D}} f_i r_i + \sum_{\{i,j\}\in\mathcal{E}} f_{ij} m_{ij}$$
$$s.t.\ (2)-(14),\ r_i, m_{ij} \ge 0,$$

where we relax the integrality requirements of variables r_i and m_{ij} for a faster solution. Letting W^{LP} denote the optimal solution of **RP4**, the following inequality is also a valid cut for **P**:

$$\sum_{i\in\mathcal{I}\setminus\mathcal{D}} f_i r_i + \sum_{\{i,j\}\in\mathcal{E}} f_{ij} m_{ij} \ge W^{LP} \quad \text{where} \quad r_i, m_{ij} \ge 0. \quad (47)$$

5.2.4.3. Minimum Required Supply Amount. Given our assumption that there are no shortages in meeting demands, it is necessary to establish DCs capable of fulfilling all demands. For the given quantities at each shelter, we must open DCs that can provide at least the minimum required supply for all shelters. In the overall model **P**, this consideration is addressed through Constraints (4)–(8), which are now incorporated in the subproblem. Thus, to ensure a feasible solution for the original problem derived from the master problem, we introduce the following surrogate constraint:

$$\sum_{o\in\mathcal{O}} r_o s_{og} \ge w_g \sum_{d\in\mathcal{D}} s_d \qquad \forall g \in \mathcal{G}. \quad (48)$$

Constraints (48) ensure that we open a sufficient number of distribution centers to meet all the available demands of shelters for each item g.

5.2.4.4. Minimum Number of Arcs for Paths. As the solution of the master problem plays a critical role in ensuring the feasibility of the subproblem, we are motivated to obtain an **MP** solution that supports this end by introducing a sufficient number of arcs into the **SP** network without potentially violating the lower bounding characteristic of the master problem. To achieve this, we introduce a lower bound on the number of arcs in the path for each origin–destination ($o-d$) pair.

Let k_{od} denote the number of arcs in the shortest path from a potential origin o to destination d in the original given network. Considering a unit length for each arc in that network ($\forall (i,j)\in\mathcal{A}$), we find the shortest paths for all node pairs $o-d$ that utilize the arcs in $\mathcal{O}\times\mathcal{N}, \mathcal{N}\times\mathcal{N}, \mathcal{N}\times\mathcal{D}$, and $\mathcal{D}\times\mathcal{D}$ and record their k_{od} values a priori to solving the overall problem. These k_{od} values are the lower bounds on the number of arcs for each corresponding potential $o-d$ path in the final solution. To formalize this relationship, let w_{od} be a binary variable indicating whether there is an active transfer from a DC at o to a shelter d. Then, the arc selections (m_{ij}) in the **MP** should satisfy the constraints that establish k_{od} as a valid lower bound for each active $o-d$ pair as follows:

$$y_{odg} \le 1 - w_{od} \qquad \forall o, d, g, \quad (49)$$

$$k_{od} - \sum_{(i_1,j_1)\in\mathcal{E}_1} m_{i_1 j_1} - \sum_{(i_2,j_2)\in\mathcal{E}_2} m_{i_2 j_2} - \sum_{(i_3,j_3)\in\mathcal{E}_3} m_{i_3 j_3} - \sum_{(i_4,j_4)\in\mathcal{E}_4} m_{i_4 j_4} \le M w_{od} \qquad \forall o, d, \quad (50)$$

where $\mathcal{E}_1 = \{(i_1, j_1) : i_1 \in \mathcal{O}, j_1 \in \mathcal{N}\}$, $\mathcal{E}_2 = \{i_2 \in \mathcal{N}, j_2 \in \mathcal{N}\}$, $\mathcal{E}_3 = \{i_3 \in \mathcal{N}, j_3 \in \mathcal{D}\}$, and $\mathcal{E}_4 = \{(i_4 \in \mathcal{D}, j_4 \in \mathcal{D}\}$. If there is an active assignment between o and d, Constraint (49) ensures that w_{od} is set to zero, indicating the presence of a valid path. In this case, Constraint (50) enforces k_{od} as a lower bound on the number of edges in that $o-d$ path. The left-hand side of this inequality sums the arc usage variables m_{ij} across all relevant edges, ensuring that the path includes at least k_{od} arcs. If there is no transfer from o to d, w_{od} can take the value of one, effectively relaxing the constraint.

5.3. Overall Approach

Algorithm 2 outlines the steps of our overall algorithm. First, we relax the Capacity Constraints (9) of the original problem **P** and solve it to obtain an underlying network. Second, we fix the variables for selected (opened) locations to one and set the others as free variables in **RMP** before we introduce our surrogate constraints and solve **RMP** to obtain a feasible solution with newly determined σ^t_{ijg} and free r_i and m_{ij} variables. Third, we use this solution as input to Algorithm 1 to check the feasibility of **SP** without solving it. Fourth, using the updated solution, we solve ***DSP***$_g$ for each item g. If unbounded, we add a feasibility cut; otherwise, we strengthen the optimality cut using the two-phase method and add it to **RMP**. Finally, we resolve **RMP** by incorporating the added feasibility and optimality cuts and repeat the process until no further feasible solutions exist for **RMP**. The optimal solution to the original problem ***P*** is given by the latest feasible solution of **RMP** thus obtained.

Algorithm 2 (Overall BD Algorithm with Enhancements)
1: **Initialize** $\epsilon = 2.0\%$, UB $= \infty$.
2: Solve the uncapacitated version of problem ***P*** (without (9))
3: Set the open node and arc variables as 1 (i.e., $r_i = 1$ and $m_{ij} = 1$) and the others as free in **RMP**
4: Add surrogate constraints (46)–(50) to **RMP**.
5: Solve **RMP** to obtain σ^t_{ijg} and the values for m_{ij} and r_i for free nodes and arcs.
6: **while RMP** has a feasible solution, **do**
7:   Call Algorithm 1
8:   **for** each item g **do**
9:     Solve **DSP**$_g$ with $\hat{\sigma}^t_{ijg}$, $\hat{m}_{ij}$, and $\hat{r}_i$.
10:     **if DSP**$_g$ is unbounded **then**

Generate a feasibility cut (34) and add it to **RMP**.
Set $D_g = \infty$.
end if
if DSP$_g$ is feasible **then**
Fix all values of μ_{og} and α_{dg}
Fix dual variables associated with $\hat{\sigma}^t_{ij} = 1$, $\hat{m}_{ij} = 1$, and $\hat{r}_o = 1$ in the dual objective.
Solve phase II of **DSP**$_g$ with fixed values to obtain the rest of the dual values
Add the strengthened optimality cut (33) to **RMP** using the final optimal dual values
Calculate D_g with the latest solution of **DSP**$_g$.
end if
end for
Set $UB_{new} = \sum_{i\in\mathcal{I}\setminus\mathcal{D}} f_i\hat{r}_i + \sum_{\{i,j\}\in\mathcal{E}} f_{ij}\hat{m}_{ij} + \sum_{g\in G} D_g$.
if $UB_{new} < UB$ **then**
$UB \leftarrow UB_{new}$
end if
Solve **RMP** and update $\hat{\sigma}^t_{ijg}$, $\hat{m}_{ij}$, and $\hat{r}_i$.
end while
Return UB and its corresponding solution.

6. Computational Study

To evaluate the performance of our proposed solution methodology, we design nine computational experiments across three network size groups and eight problem classes (C1–C8) as summarized in Online Table 1. The experiments systematically vary algorithmic components, including B&C, automated Benders decomposition, and our custom BD framework, to isolate and quantify the contribution of each enhancement: the max-flow-based feasibility check and arc generation (Algorithm 1), subproblem disaggregation by commodity, surrogate constraints, cut strengthening, and the ϵ-optimal method. For each experiment, we generate 10 random instances per class with varying capacities, distances, and travel times and report average runtimes and optimality gaps. Full details of instance generation, parameter distributions, and experimental setup are provided in Online Section S.2.1.

The results, reported in Online Tables 3 and 4, demonstrate that the standard B&C approach fails to solve instances of class C3 and beyond within the allotted time. Automated Benders decomposition improves upon B&C considerably, and incorporating surrogate constraints further extends its reach to class C6. Our custom BD framework with max-flow-based feasibility support (experiment 5) achieves the largest single performance gain, solving all eight classes including the largest C7 and C8 instances, underscoring that maintaining subproblem feasibility through network connectivity is the central challenge for BD on time-expanded relief networks. Combining all enhancement methods (experiment 9) yields the best overall runtime across classes, and this combined approach is, therefore, adopted for the case study in the following section.

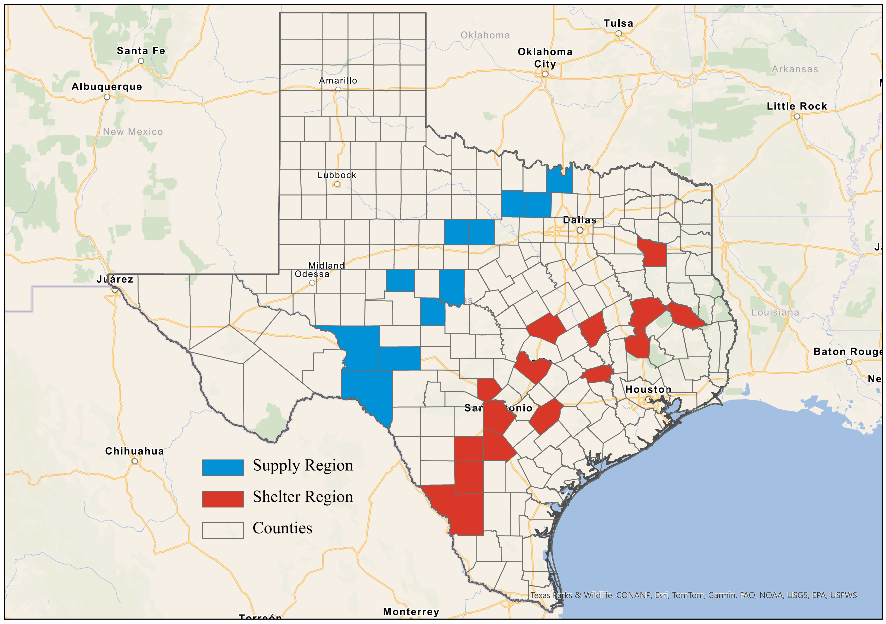
7. Case Study

This section demonstrates our proposed model through a case study using realistic GIS data from the coastal Texas area. We describe our data collection method and present the base solution to illustrate the model's ability to generate feasible and cost-effective relief distribution plans under realistic network and demand conditions. Sensitivity analyses examining the effects of key supply distribution parameters and alternative sheltering configurations on the optimal solution are provided in Online Section S.3.

7.1. Data Collection

We consider a study region, Figure 3, including potential supply and shelter regions based on the counties and their population, and evacuation routes provided by the Texas Department of Transportation (2024). Figure 3 shows the complete study region with 11 counties in our relief supply region (north central Texas, blue) and the 15 counties in our designated shelter regions (south Texas, red). Rather than individual infrastructure, we identify counties as shelters; however, within each county, we consider public schools with known shelter IDs (National Center for Education Statistics 2024) and more than 1,500 enrollment as potential shelters. We then sort these counties by their number of potential shelters and select the top 15 counties as our final shelter locations. Within each county, all schools that serve as shelters are aggregated into a single super shelter node with capacity equal to the sum of the school capacities in that county, and deliveries and assignments are made to this super shelter, whereas intracounty transfers are assumed negligible relative to intercounty travel. As for the capacity of a shelter county, we assume that about 12% of its schools' enrollment identify as shelters. For the case study, the total system demand is approximately 55,000 shipment units (evacuation population at shelters multiplied by the number of item types and the per-person requirement for each item, aggregated across all shelters). The demand level is scaled to fit within a 15-period operational horizon, preserving the relative demand-to-capacity structure of the region. Infrastructure capacities are derived from lane information and converted into shipment throughput using an operational allocation parameter, ensuring that routing behavior and bottleneck formation remain representative of larger scale scenarios. This scaling allows the model to capture intraday routing dynamics at hourly resolution, maintaining computational tractability.

Figure 3. (Color online) Texas Shelter and Relief Regions



For DCs, we select the counties with a population more than 3,000 in north central Texas (Texas Demographics Center 2024); the selected supply and shelter counties and their population levels are reported in Table 2. We assign capacities for each type of item in a way that ensures enough supply for all evacuees under the worst case scenario. Each eligible county is represented by a single DC that stands in for one or more physical sites; its capacity is the aggregate across those sites and is sized to cover the worst case evacuee count with a small safety margin.

In the study road network depicted in Figure 4(a), all intersection points can act as transfer nodes for relief items to be transferred to the shelter destinations. We determine actual distances between nodes and estimate travel time between them based on the average speed limit on the corresponding road types, including interstate, principal arterial, minor arterial, major collector, and minor collector roads (Texas Department of Transportation 2024). Specifically, we calculate the upper distance (UD_{ij}) for each arc using the formulation detailed in Online Section S.2.1, in which the base distance d_{ij} corresponds to the actual physical distance obtained from GIS data. To convert this operational distance into time, we divide the calculated UD_{ij} by the average speed limit associated with the road type (e.g., 65 miles per hour (mph) for interstates). The resulting values are then normalized relative to the minimum travel time in the network, approximately 60 minutes given the geographic scale, and rounded to the nearest integer. Thus, the discrete time unit in our model represents one hour.

We determine outflow capacities from the nodes based on the number of lanes. Arc capacity q_{ij} represents the maximum relief flow entering arc (i, j) per hour, computed as the product of usable lanes and an operational throughput parameter. The operational throughput parameter represents the portion of capacity allocated to relief movements when the network remains shared with emergency vehicles and background traffic (Haghani and Oh 1996; Özdamar, Ekinci, and Küçükyazici 2004).

As a plausibility check, we compare total shipment demand with the aggregate lane-hours available across

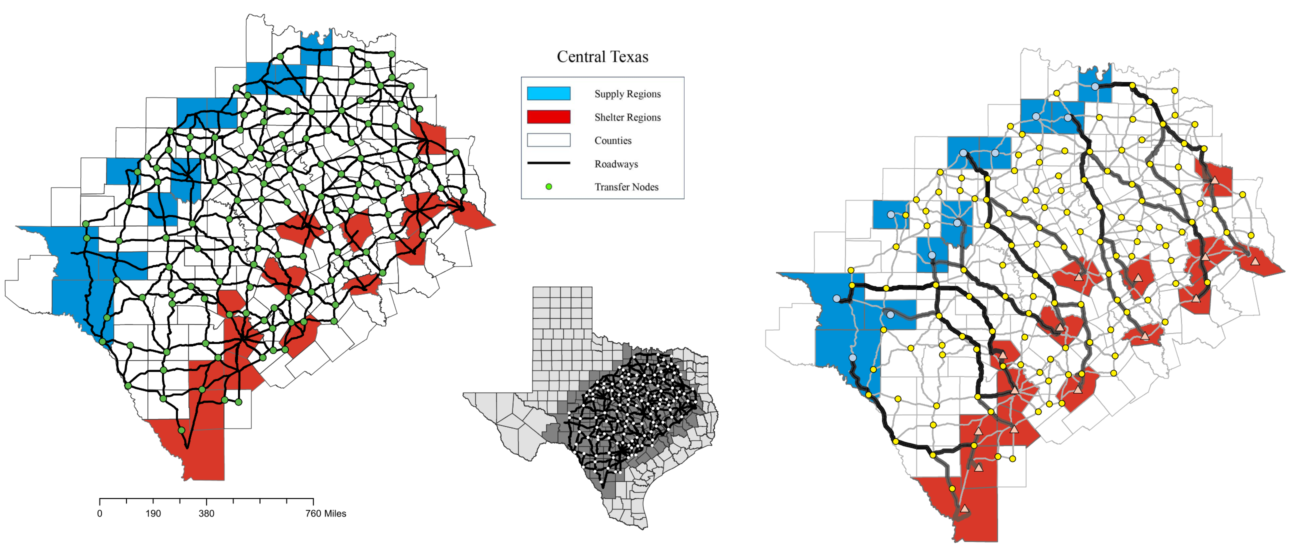
Table 2. Texas Relief and Sheltering County Population Data

	Supply regions			Shelter regions	
	County	Population		County	Population
1.	Coke	3,285	1.	Angelina	86,396
2.	Coleman	7,684	2.	Atascosa	48,981
3.	Concho	3,303	3.	Bell	370,647
4.	Cooke	41,668	4.	Bexar	2,009,324
5.	Crockett	3,098	5.	Frio	18,385
6.	Jack	8,472	6.	Gonzales	19,653
7.	Shackelford	3,105	7.	Houston	22,066
8.	Stephens	9,101	8.	Kendall	44,279
9.	Sutton	3,372	9.	La Salle	50,088
10.	Val Verde	47,586	10.	Robertson	16,757
11.	Wise	68,632	11.	Smith	233,479
			12.	Travis	1,290,188
			13.	Walker	76,400
			14.	Washington	35,805
			15.	Webb	267,114

Figure 4. (Color online) Case Study Region

(a) Locations of Shelters and Relief, and the Road Network

(b) Case study optimal solution



the primary response corridors connecting DC and shelter regions in the extracted GIS network. These corridors consist of 25 interregional arcs identified from the case study network. The average number of usable lanes, equal to 2.9, is computed directly from Texas Department of Transportation lane attribute data across these arcs, which include two-, three-, and four-lane roads. Given the dispatch cutoff time and arc travel durations, each such arc remains operational for approximately 11 hourly periods within the 15-period planning horizon based on the earliest feasible departure time at its origin and the latest admissible arrival time at its destination. Under these network characteristics, the implied throughput required to clear total demand is approximately 69 shipment units per lane per hour. The adopted baseline value of 75 shipment units per lane per hour, therefore, ensures feasibility within the operational horizon, remaining conservative relative to physical roadway capacity because only a fraction of total lane capacity can be allocated to organized relief movements during response operations. Sensitivity analysis further evaluates scenarios in which this operational throughput allocation is restricted.

To clarify parameter construction, consider an arc (i, j) connecting transfer nodes. Suppose the roadway segment has three usable lanes and the operational throughput parameter is 75 shipment units per lane per hour. Then the per-period arc capacity is $q_{ij} = 3 \times 75 = 225$ shipment units.

Baseline travel times are obtained through a preprocessing step using the upper distance parameter $UD_{ij} = d_{ij} + \frac{freq_{ij}}{q_{ij}} L$, where d_{ij} is the GIS-based physical distance, $freq_{ij}$ measures structural importance in shortest paths, and L is a scaling constant. For illustration, let $d_{ij} = 78$ miles, $freq_{ij} = 10$, $L = 500$, and $q_{ij} = 225$. Then, $UD_{ij} = 78 + \frac{10}{225} \times 500 = 100.22$. Dividing by an average speed of 65 mph yields a continuous travel time of approximately 1.54 hours. After normalization and discretization, this results in $t_{ij} = 2$ periods. Thus, flow entering node i at time t reaches node j at time $t + 2$ in the time-expanded network.

Travel times remain fixed during optimization. Capacity q_{ij} acts solely as a per-period throughput constraint, and its role in preprocessing is to differentiate structural travel conditions across arcs without introducing flow-dependent congestion effects.

The variable cost for unit flow of each item on the road segments is estimated by considering loading/unloading costs, fuel costs, and driver salaries. Furthermore, the associated costs of nodes are calculated as a function of their population and the number of nodes in each county. This reflects the increased coordination effort and operational complexity typically encountered in more densely populated areas, consistent with the nodal processing bottlenecks inherent in disaster response.

The construction of model inputs ensures consistency between the numerical experiments and the case study. Travel times and arc capacities are calibrated to preserve the structural stress of the system, allowing the exact optimization of the 55,000-demand instance to provide rigorous insights into the bottleneck behavior and routing strategies required for larger populations. Furthermore, this hourly resolution is specifically selected to capture the high-frequency decision-making characteristic of the initial response phase. Whereas modeling the full regional population in a single instance would require redefining the time unit from hours to days to maintain tractability, such aggregation would obscure critical intraday routing dynamics. Instead, this proposed framework is designed to be applied iteratively—managing the total demand through sequential daily operational windows—thereby preserving both computational feasibility and operational precision. All relevant case data and details of calculations can be found at https://s2.smu.edu/INETS/TestInstances/ReliefND-VR1.htm.

7.2. Base Solution to Relief Distribution

Taking the initial parameter values as set above, we solve our model to obtain a base or benchmark solution to compare the performance of the parametric variations. The resulting network for the base problem is presented in Figure 4(b). This solution chooses 9 out of 11 DCs to satisfy the shelter's relief supply demands. Routes used are depicted with thicker and gradually darker segments as the frequency of their usage increases. The underlying road network is also shown in thin light gray. We observe that each DC serves multiple shelters, and each shelter receives supplies from several DCs.

Furthermore, to highlight instances of supply flow in the network and for further model validation and verification, we display the active arcs with ongoing flows for each time unit from start to finish in Figure 5. This easily constructed flow visualization aids decision makers in understanding the flow dynamics within the network at different times, thus enabling the adaptability of decisions during unpredictable events. Departure of supply relief from the DCs begins at the onset of the planning horizon ($t = 0$) and continues until the end of the allowed time ($T_o = 5$). Over time, the activated arcs channel flow toward shelters with the final deliveries arriving at $t = 13$. Traffic flows are predominantly higher on arcs close to the DCs as a significant volume of relief items initially moves through these arcs. These arcs receive flows from multiple DCs and subsequently diverge to meet the needs of various shelters toward which the relief flows from multiple DCs converge.

Sensitivity analyses examining the effects of key supply distribution parameters and alternative sheltering configurations are provided in Online Section S.3. Eight scenarios obtained by varying unit flow cost at transfer nodes (u_i), dispatch cutoff time (T_o), and per-period arc throughput capacity (q_{ij}) across high and low settings demonstrate that tighter dispatch windows and reduced arc throughput trigger activation of additional DCs and arcs, whereas higher node-associated costs shift routing toward corridors that bypass densely populated transfer locations. Additionally, increasing the number of shelter locations, holding total system demand fixed, shows that a more spatially distributed shelter configuration reduces total system cost by 12% relative to the base solution.

8. Concluding Remarks

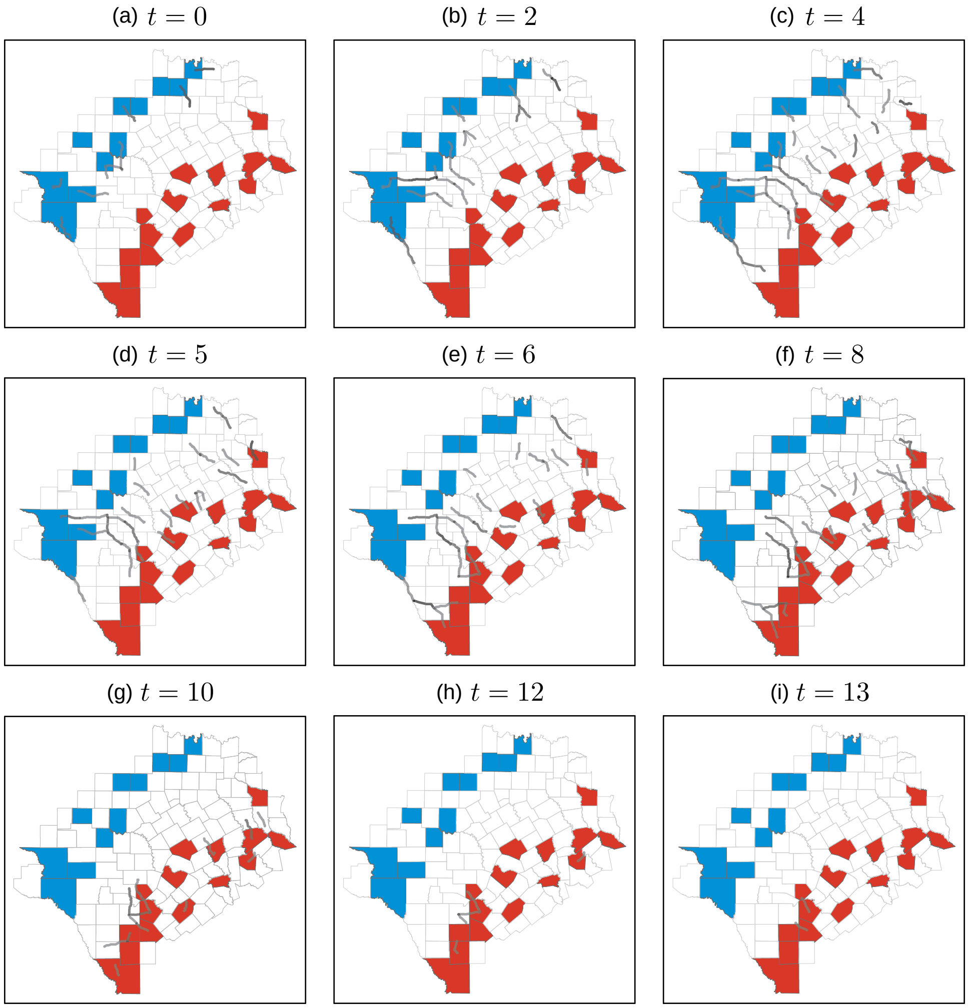
This study introduces an emergency relief network design problem for disaster response, focusing on cost-effective, tactical decision making for the supply side of evacuation plans. Considering cost factors, we construct a time-expanded network to address key tactical questions.

We develop a new mixed-integer linear programming model that incorporates fixed facility and network utilization charges, transportation costs, and operational cost components associated with routing and transfer activities as objectives. The model operates under time, flow balance, capacity, and demand satisfaction constraints, providing a detailed tactical relief distribution plan from distribution centers to shelter evacuees. Whereas a state-of-the-art B&C approach provides accurate solutions, it proves computationally expensive for large-scale instances. To address this, we propose a BD-based algorithm and evaluate three enhanced implementations through a comprehensive computational study.

We apply the model to solve a realistic problem using publicly available GIS-based data, demonstrating its potential as a tactical decision-making tool for emergency managers. The model provides insights into the dynamics of relief distribution during disaster scenarios and explores the effects of variables such as allowable distribution time, arc capacities, and operational routing and capacity parameters. Additionally, we compare two sheltering strategies: one with a greater number of smaller shelters with lower fixed costs and another with fewer, larger shelters with higher fixed costs. We observe that increasing the number of shelters reduces the overall system cost.

In the algorithmic context, Benders decomposition is a powerful technique as an optimization methodology framework; however, it frequently requires enhancements that are also supported for implementation because of the framework itself following a clear separation of bound problems. All of the acceleration methods that are included in Section 5.2 have the potential to be utilized in similar network design problems, both capacitated and uncapacitated and both time-phased or single period. We especially emphasize the difficulties that motivate our enhancements as they are likely to be seen in other settings individually or in some combination. Specifically, in the BD framework, as some of the information in the overall problem is moved to the subproblem, the master problem loses its handle on ensuring flow feasibility when making the (location and arc) choice decisions to determine the network. This frequently leads to a disconnected

Figure 5. (Color online) Case Study Optimal Solution



underlying network coupled with inadequate arc capacities provided by the MP solution and, eventually, to a flow subproblem that is infeasible. Similar situations arise in multicommodity flow network design problems and multiproduct capacitated production-distribution network design problems among many others. Adding the time dimension to flows complicates these issues even further. Especially, the enhancements that we discuss under Sections 5.2.1 and 5.2.4 can be further extended to address the difficulties that arise in adopting Benders decomposition in these settings on time-phased networks. In (36), we establish a relationship between the time component and the arc capacities in the network and obtain an approximation to generate zero residual arcs by solving a maximum flow problem. This type of relationship is general enough to help establish similar relationships in those time-phased network design settings to improve computational performance. We provide our algorithm for this purpose explicitly in Algorithm 1 to support these extensions. Likewise, the surrogate constraints, especially the ones that utilize partial objective values and minimum required number of arcs and supply amounts are general enough to be utilized in those network design settings that seek optimal paths under capacity and time constraints. The other acceleration approaches, specifically the cut-strengthening via updating dual variable values associated with zero-valued primal solutions and the ϵ-optimal method, are employed with mixed success in the literature; however, they show promise in this context as presented in our experiments.

This study highlights the importance of time-expanded network design in disaster relief, enabling decision makers to coordinate supply distribution over multiple periods for faster and more efficient aid delivery. Unlike static models, this approach helps anticipate operational bottlenecks arising from limited throughput and time availability, optimize routing schedules, and allocate resources effectively. By incorporating costs, transportation constraints, and sheltering strategies, the model is a decision-support tool for emergency planners, guiding supply prepositioning and real-time response adjustments. This framework enhances logistical efficiency, reduces delays, and improves resource utilization, supporting faster recovery and minimizing human suffering during disasters.

This study can be extended by integrating evacuation and relief logistics into a unified model to examine their interactions and effects on tactical decision making. Whereas this research adopts a deterministic approach, incorporating uncertainties such as network disruptions, variations in evacuee numbers, and destination preferences would provide a more robust framework for disaster response planning. It would also be valuable to model the behavioral decisions of evacuees, including partial compliance with assigned shelters, endogenous destination choice as information evolves, and spillover to neighboring counties, and these can be represented with gravity models, discrete choice, or simple compliance rates that feed the shelter demand process. Traffic effects can be enriched in future work by introducing link-level dynamics such as time-dependent speeds and capacities, queue formation and spillback, lane drops, contraflow operations, incident clearance times, and weather-related degradation. These extensions would allow explicit representation of traffic dynamics beyond the fixed travel-time and capacity assumptions adopted in the present study and represent promising directions for future research.

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Endnote

[1] We note that the destination points do not need to be shelters specifically; any other location expected and/or designed to receive a significant intake of evacuees can be considered a demand location as well.

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